FF-S-740a
October 8, 1965

SUPERSEDING
Int. Fed. Spec. FF-S-00740 (GSA-FSS)
July 22, 1964, and
Fed. Spec. FF-W-556
May 1, 1934

FEDERAL SPECIFICATION

STEEL WOOL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICAION

1.1 Scope. Steel wool covered by this specification is used as an abrasive for cleaning and polishing (see 6.1).

1.2 Classification. Steel wool shall be furnished in the following types, classes and forms as specified (see 6.2):

Types

Type I—Carbon steel
Type II—Ferritic stainless steel

Classes

Type I
0000 fine
000
00
0
1
2
3 Coarse

Classes

Type II
- -
- -
- -
- -
1
2
3

Forms

Types I and II—rolls
pads

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-B-636—Box, Fiberboard

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Informa- Standards and at the prices indicated in the Index. tion in the Index of Federal Specifications and The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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TABLE I. *Thickness requirements for individual fibers*

Class	Not more than 5 percent under	Mean width of fiber	Not more than 5 percent over	No fiber to exceed
	Inch	*Inch*	*Inch*	*Inch*
0000	—	0.0006—0.0010	0.003	0.006
000	—	.0008— .0013	.004	.008
00	—	.0012— .0022	.005	.010
0	0.0003	.0016— .0024	.007	.012
1	0.0005	.0018— .0026	.010	.015
2	0.001	.0040— .0060	.014	.025
3	.002	.0070— .0095	.018	.040

books and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Type I steel wool shall be produced from plain carbon steel and type II from a ferritic grade of stainless steel such as AISI type 430.

3.2 Dimensions and weights.

3.2.1 *Rolls.* Weights of each steel wool roll shall be one pound, minimum. Length of rolls shall be 14 and 16 inches long.

3.2.2 *Pads.* Weight of each steel wool pad shall be 3/4 ounce, minimum. Pads shall be formed from not less than 3-3/4 inch wide ribbon.

3.3 Fibers (types I and II).

3.3.1 *Edge of fibers.* Individual fibers shall have uniform sharp cutting edges (see 4.3.1).

3.3.2 *Length.* The length of fibers shall be such that wool will cling together in handling without excessive unraveling.

3.3.3 *Thickness.* Unless otherwise specified (see 6.2), the thickness of individual fibers shall be in accordance with table I (see 4.3.2).

TABLE II. *Size of mandrel for ductility requirements*

Class	Diameter of mandrel
	Inch
00, 000, 0000	0.028
1 and 2	.036
2 and 3	.045

3.3.4 *Ductility.* Individual fibers of types I and II steel wool shall be capable of being bent completely around a wire mandrel of the size listed for grades of steel wool in accordance with table II. Tolerance on the diameter of mandrel shall be plus or minus 0.002 inch (see 4.4.2).

3.4 Wool.

3.4.1 *Corrosion resistance.*

3.4.1.1 *Type I.* Carbon steel wool shall not discolor the blotting paper under wool specimen when tested in accordance with 4.4.3.1.

3.4.1.2 *Type II.* Ferritic stainless steel wool shall not exhibit any evidence of corrosion attack when tested in accordance with 4.4.3.2.

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3.4.2 *Moisture and oil content.* Unless otherwise specified (see 6.2), type I wool shall not contain a total of moisture and oil greater than 1.5 percent by weight (see 4.4.4).

3.4.3 *Elasticity.* Both types of wool shall be capable of springing back toward their original shape when tested in accordance with 4.4.5.

3.5 Workmanship. Steel wool shall be of uniform quality, clean, and free from foreign material, rust, chips, and defects which might render the material unsuitable for the purposes intended.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 *Lot.* Unless otherwise specified (see 6.2), a lot shall consist of all wool of the same type and class submitted for inspection at one time.

4.2.2 *Sampling for examination of packaging and packing.* Packed wool, ready for shipment, shall be sampled in accordance with level I.

4.2.3 *Sampling for dimensional and visual examination of end item.* Sampling of wool for these characteristics shall be in accordance with inspection level S-4.

4.2.4 *Sampling for tests.* Samples for tests shall be selected in accordance with inspection level S-2. Samples for test may be obtained from those selected in 4.2.3.

4.3 Examination. Packed shipping containers and wool selected in 4.2.2 and 4.2.3 shall be examined in accordance with table II. The AQL's for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects in accordance with MIL-STD-105.

4.3.1 *Edge of fibers.* The edge of fibers shall be examined by not less than 30x magnification (see 3.3.1).

4.3.2 *Thickness of fibers.* Twenty fibers shall be measured for maximum lateral dimension under not less than a 50x microscope fitted with a recticle suitable for reading the fibers width in thousandths of an inch (see 3.3.3).

4.3.3 *Extraneous material in wool.* Wool shall be examined at 4x or greater magnification for the presence of foreign material and chips (see 3.5).

4.3.4 *Compactness of wool.* Examine wool after handling to ascertain whether the wool clings together without excessive unravelling (see 3.3.2).

TABLE III. *Classification of defects*

Examine applicable paragraph	Defects	Major	Minor
End item fibers			
3.3.1	Edge of fibers not sharp	X	
Wool			
3.5	Foreign material in wool		X
3.5	Wool stained with rust		X
Packaging			
5.1	Wool not wrapped in chipboard or sleeve		X
5.1	Chipboard not correct design		X

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TABLE III. *Classification of defects (cont'd)*

Examine applicable paragraph	Defects	Major	Minor
5.3.1, 5.3.2.1	Unit packages not marked as required		X
3.2	Unit packages not correct weight		X
Shipping containers			
5.2	Inadequate closure		X
5.2	Tape missing or not reinforced as specified		X
5.2	Not strapped, when required		X
5.2	Score lines broken or torn		X
5.3	Marking missing, incorrect or illegible		X

4.4 Tests.

4.4.1 *AQL.* The AQL for tests shall be 6.5 percent on the basis of percent defective.

4.4.2 *Ductility.* A single fiber of each type and grade of wool shall be wound, in a single loop, tightly around a taut wire. When necessary, retest of fibers shall be in accordance with the general section of Fed. Test Method Std. No. 151 (see 3.3.4).

4.4.3 *Corrosion.*

4.4.3.1 *Type I.* A tuft of steel wool shall be lightly compressed in a ball shape, about 3/4 inch diameter, and then slightly flattened. Dip a small sheet of white blotting paper in distilled water, remove from water, shake off excess water and place at the bottom of a glass container. Place the wool specimen on top of the blotting paper and cover the container. Temperature of the container shall be maintained at 72° ± 5° F. for a period of 5 hours (see 3.4.1.1).

4.4.3.2 *Type II.* Prepare a nitric acid solution by adding 5 ml. concentrated nitric acid (sp. gr. 1.42) to 95 ml. distilled water. Drop two drops of this solution in the center of a balled and compressed tuft of the steel wool to determine compliance with 3.4.1.2.

4.4.4 *Moisture and oil content.* When required, the loss of weight of moisture and oil shall be determined using a 50 gram sample of wool dried in an oven at 250° F. for 1 hour (see 3.4.2).

4.4.5 *Elasticity.* Measure the height of a tuft, about 3/4 to 1 inch average height, or a pad of type I or II wool in its natural state. Compress the wool to 50 percent of the original height, allow wool to spring toward its original shape. The wool shall spring back to within 85 percent of the original measurement (see 3.4.3).

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packaging. Packaging shall be class A or C as specified (see 6.2).

5.1.1 *Level A (types I and II).*

5.1.1.1 *Rolls.* One pound of steel wool shall be packaged in a chipboard paper sleeve approximately 4 inches in diameter and 14 to 16 inches long.

5.1.1.2 *Pads.* Sixteen 3/4 ounce steel wool pads shall be packaged in a chipboard paper sleeve measuring 14 to 16 inches long, 4 to 6 inches wide and 2 to 4 inches high.

5.1.2 *Level C (type I or II, rolls or pads).* Packaging shall be in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be class A, B, or C as specified (see 6.2).

5.2.1 *Level A (type I or II, rolls and pads).* Twenty-five unit packages shall be packed in a snug fiberboard box conforming to PPP-B-636, V3c or V3s. The boxes shall be closed, waterproof sealed with tape and strapped in accordance with the appendix

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to the box specification, except that flat steel strapping shall be type I, class B.

5.2.2 *Level B (type I or II, rolls or pads).* Twenty-five unit packages shall be packed in a snug fitting fiberboard box conforming to PPP-B-636, type CF or SF, class domestic. Box closures, sealing and strapping shall conform to the appendix of the box specification.

5.2.3 *Level C (type I or II, rolls or pads).* Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Unit packages.* Unit packages marked or labeled in accordance with standard commercial practice will be acceptable. Marking shall include the applicable Federal Stock Number.

5.3.2 *Civil agencies.* In addition to any special marking required in the contract or order, unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 including specification classification (type, class, and form).

5.3.3 *Military activities.* In addition to any special marking required in the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Steel wool is used as an abrasive material. Type II wool is used for filtering liquids. Type II stainless steel wool should be used on stainless steels, nonferrous metals and other surfaces subject to atmospheric or other corrosive media where the use of type I wool might result in rust staining by particles of wool left on the surface. Some typical uses of steel wool are as follows:

6.1.1 *Class* 0000. Finishing and polishing fine furniture, rubbing down final varnish, lacquer and shellac finishes.

6.1.2 *Class* 000. Rubbing down final coat, dulling varnish and antique finishing.

6.1.3 *Class* 00. Also used for rubbing down final coat for cabinet work, and smoothing automobile bodies for paint spraying.

6.1.4 *Class* 0. Polishing and cleaning metal, finishing and reviving old painted surfaces and removing discoloration stains.

6.1.5 *Class* 1. Preparation of surfaces for first coats, polishing sheet metals and machinery parts, and smoothing woodwork.

6.1.6 *Class* 2. Floor maintenance, cleaning linoleum, scrubbing floors, cleaning machinery and tools, removing rust, preparing surfaces before painting or varnishing first coats and finished coats.

6.1.7 *Class* 3. Removing old paint, shellac, varnish, and rust, smoothing rough surfaces and cleaning floors.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and form required (1.2 and 6.2).
(*c*) Thickness requirements for individual fibers, if other than those specified in table I (3.3.3 and 4.3.2).
(*d*) Moisture and oil content required for type I wool if other than that specified in paragraph 3.4.2.
(*e*) Size of lot, if different from 4.2.1.
(*f*) Applicable level of packaging and packing required (5.1 and 5.2).
(*g*) Special marking if required (5.3).

6.3 **Transportation description.** The transportation description and minimum weights applicable to this commodity are:

Types I and II

Rail:

Carload minimum weight 16,000

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pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Truckload minimum weight 16,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.4 Other metal wool. Steel wool, type I, is also available in 4 inch wide ribbons wound in spools, five or six 5-pound spools per box, four 12-1/2 pound spools per box, and up to 8 catchweight reels (average 13-15 pounds per spool) per box.

6.5 Civil agency procurement. When level B packaging is specified for civil agency procurement the requirements in 5.1.1 level A packaging shall apply.

MILITARY CUSTODIANS:

Army—MR

Navy—YD

Air Force—69

Review activities:

Army—MR, WC

Navy—YD, SH

Air Force—69, 85

User activities:

Army—MI, MO, MU

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